Practical no. 16

Aim - Page Object Model (POM) Framework Implementation

Design and implement automation framework using Page Object Model (POM). Create reusable page classes and execute modular test cases.

Step 1: Create Maven Project

In Eclipse: File → New → Maven Project

Select: Create a simple project

Group Id: com.selenium

Artifact Id: POMFramework

Version: 0.0.1-SNAPSHOT

Packaging: jar

Click Finish.

Step 2: Add Selenium and TestNG Dependencies

Open pom.xml and replace its contents with:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.selenium</groupId>
    <artifactId>POMFramework</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <properties>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <maven.compiler.source>11</maven.compiler.source>
        <maven.compiler.target>11</maven.compiler.target>
    </properties>
    <dependencies>
        <!-- Selenium WebDriver -->
        <dependency>
            <groupId>org.seleniumhq.selenium</groupId>
            <artifactId>selenium-java</artifactId>
            <version>4.35.0</version>
        </dependency>
        <!-- TestNG -->
        <dependency>
            <groupId>org.testng</groupId>
            <artifactId>testng</artifactId>
            <version>7.11.0</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
</project>
```

Save with Ctrl + S

Then: Right-click project → Maven → Update Project → select project → OK

Wait for Maven to download the dependencies.

Step 3: Create Login Page Class

Right-click src/main/java → New → Package → Enter: pages

Right-click pages → New → Class → Class name: LoginPage

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
```

```
public class LoginPage {

    WebDriver driver;

    // Locators
    By username = By.id("user-name");
    By password = By.id("password");
    By loginButton = By.id("login-button");

    // Constructor
    public LoginPage(WebDriver driver) {
        this.driver = driver;
    }

    // Enter username
    public void enterUsername(String user) {
        driver.findElement(username).sendKeys(user);
    }

    // Enter password
    public void enterPassword(String pass) {
        driver.findElement(password).sendKeys(pass);
    }

    // Click login button
    public void clickLogin() {
        driver.findElement(loginButton).click();
    }

    // Complete login operation
    public void login(String user, String pass) {
        enterUsername(user);
        enterPassword(pass);
        clickLogin();
    }
}
```

Step 4: Create Home Page Class

Right-click pages → New → Class → Name: HomePage

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class HomePage {

    WebDriver driver;
    WebDriverWait wait;

    // Locator for Products heading
    By productsTitle = By.className("title");

    // Constructor
    public HomePage(WebDriver driver) {
        this.driver = driver;
        this.wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    }

    // Get page title
    public String getPageTitle() {
        return wait.until(
            ExpectedConditions.visibilityOfElementLocated(productsTitle)
        ).getText();
    }

    // Check whether Home page is displayed
    public boolean isHomePageDisplayed() {
        return wait.until(
```

```
            ExpectedConditions.visibilityOfElementLocated(productsTitle)
        ).isDisplayed();
    }
}
```

Step 5: Create Test Package

Right-click src/test/java → New → Package → Enter: tests

Right-click tests → New → Class → Class name: LoginTest

Step 6: Write the Test Class (LoginTest)

```
package tests;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;
import pages.HomePage;
import pages.LoginPage;

public class LoginTest {

    WebDriver driver;
    LoginPage loginPage;
    HomePage homePage;

    @BeforeMethod
    public void setUp() {
        driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.get("https://www.saucedemo.com/");
        loginPage = new LoginPage(driver);
        homePage = new HomePage(driver);
    }

    @Test
    public void validLoginTest() {
        loginPage.login("standard_user", "secret_sauce");
        Assert.assertTrue(
            homePage.isHomePageDisplayed(),
            "Home page is not displayed"
        );
        Assert.assertEquals(
            homePage.getPageTitle(),
            "Products"
        );
    }

    @AfterMethod
    public void tearDown() {
        if (driver != null) {
            driver.quit();
        }
    }
}
```

Step 7: Install TestNG Plugin in Eclipse

1. Help → Eclipse Marketplace.
2. Search for TestNG for Eclipse.
3. Click Install.
4. Accept the license and restart Eclipse when asked.
5. After restarting, right-click LoginTest.java → Run As → TestNG Test should appear.

Step 8: Run the Test

Right-click LoginTest.java → Run As → TestNG Test